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Reconciling Latent Variables and Networks

Exploring and extending the Psychometric-Toolbox

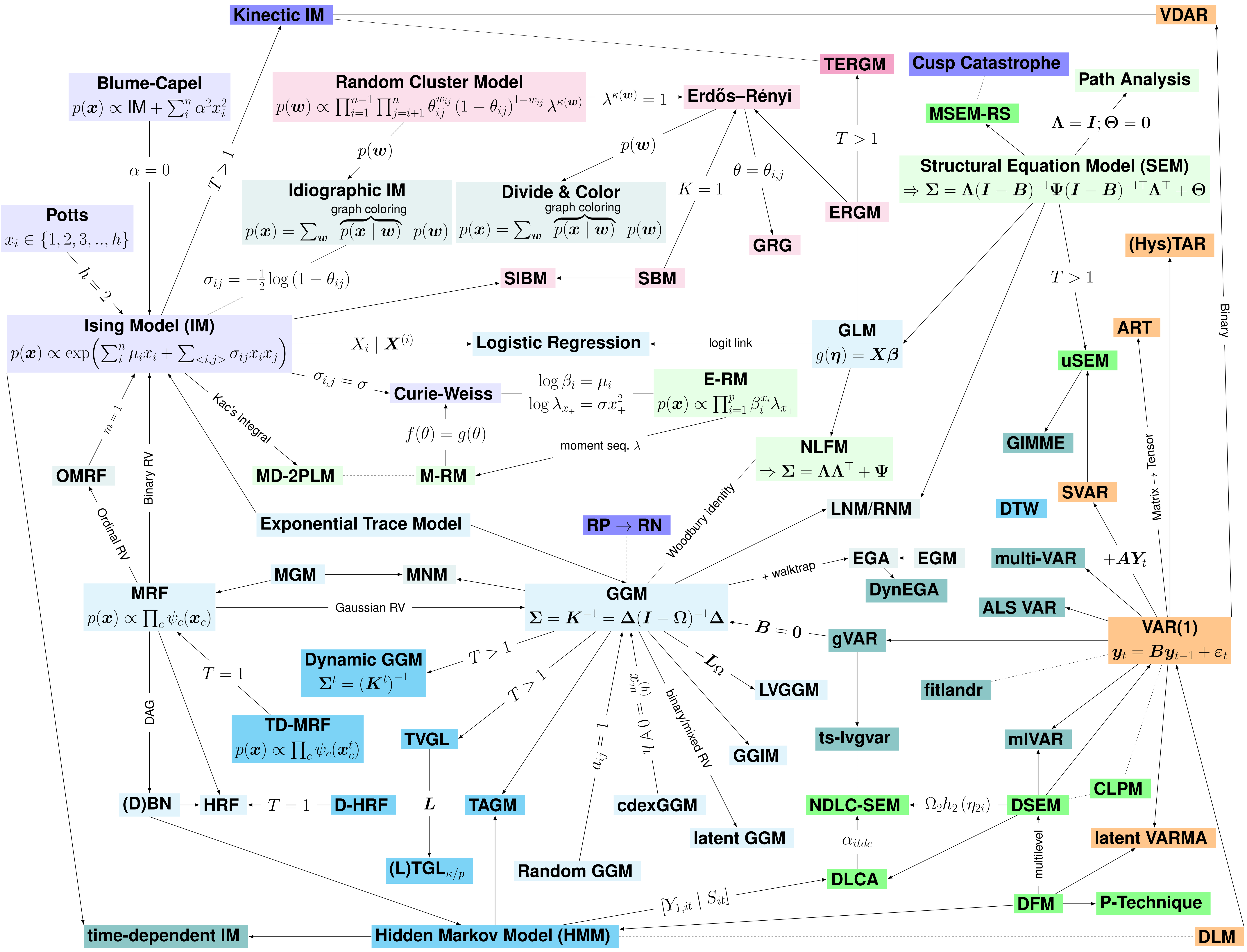

Figure 1: *With light colored models for cross-sectional and darker colored model for time-series data from: Psychometrics / ▬, Network Psychometrics / ▬, Econometrics, Statistics & Machine Learning / ▬, Graph Theory & Network Science / ▬, Physics / ▬.* ***Model Names****: Autoregressive Tensor Model (ART), Cross-Lagged Panel Model (CLPM), covariate-dependent GGM (cdexGGM), Dynamic Bayesian Network (DBN), Dynamic Factor Model (DFM), Dynamic Hybrid Random Field (D-HRF), Dynamic Linear Model (DLM), Dynamic Structural Equation Model (DSEM), Dynamic TimeWarping (DTW), Erdős–Rényi (ER or Random Graph Model), Exploratory Graph Analysis (EGA), Exploratory Graph Model (EGM), Exponential Random Graph Model (ERGM), Extended Rasch Model (E-RM), Gaussian Graphical Interaction Model (GGIM), Gaussian Graphical Model (GGM), Generalized Linear Model (GLM), Generalized Random Graph (GRG), graphical VAR (gVAR), Group Iterative Multiple Model Estimation (GIMME), Hysteretic Threshold Autoregressive (HysTAR), (HRF), Latent Network Modeling (LNM), Latent Variable Gaussian Graphical Models (LVGGM), Latent Variable Time-Varying Graphical Lasso (LTGL), Latent Variable Time-Varying Graphical Lasso with pattern inference (LTGL$_p$), Latent Variable Time-Varying Graphical Lasso with temporal kernel (LTGL$_\kappa$), Marginal Rasch Model (M-RM), Markov Random Field (MRF), Mixed Graphical Model (MGM), Mixture SEM with Regime-Switching (MSEM-RS), Moderated Network Model (MNM), Multidimensional two-parameter Logistic Model (MD-2PLM), Nonlinear Dynamic Latent Class - Structural Equation Modeling (NDLC-SEM), Ordinal Markov Random Field (OMRF), Residual Network Modeling (RNM), Structural Vector Autoregressive (SVAR), Stochastic Block Ising Model (SBIM), Stochastic Block Model (SBM), Temporal Exponential Random Graph Model (TERGM), Time Adaptive Gaussian Model (TAGM), Time-Dynamic Markov Random Field (TD-MRF), Time-Varying Graphical Lasso (TVGL), unified SEM (uSEM), Vector Autoregressive (VAR), Vector Discrete Autoregressive (VDAR)*

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